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Privacy-Preserving Methods for Data Science and Distributed Systems

University of Basel

Differentially Private Learning for Blood Cell Image Classification

Outline

Problem and Dataset: importance of blood cell image classification and privacy concerns

Differentially Private SGD and its challenges

Solution Approach: hyperparameter search on a public dataset

Results of the private model with a fixed privacy budget

Detour: could we have done better?

Conclusions and analysis of utility and privacy trade-off

Problem and Dataset

Problem

This project focuses on classifying **blood cell images** using a **convolutional neural network** (CNN) to help accelerate the diagnosis of blood diseases.

Privacy concerns

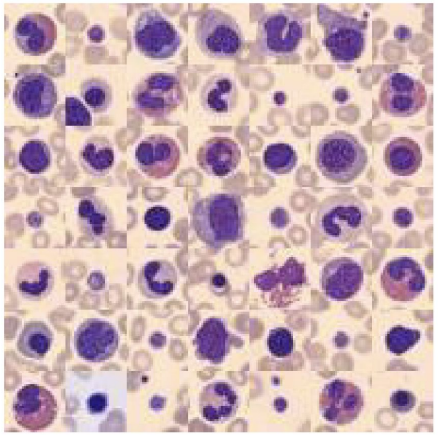
- Blood cell images may reveal rare markers that could identify patients.
- Publishing medical datasets risks non-compliance with GDPR.

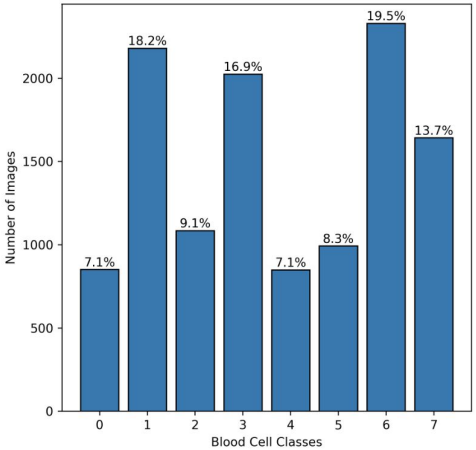
In this project, we **compare** private and non-private CNNs to study the impact of differential private learning on utility and performance for classifying blood cell images.

Dataset

BloodMNIST (from MedMNIST):

- 17,092 color images (28×28 pixels)
- 8 classes
- Treated as private



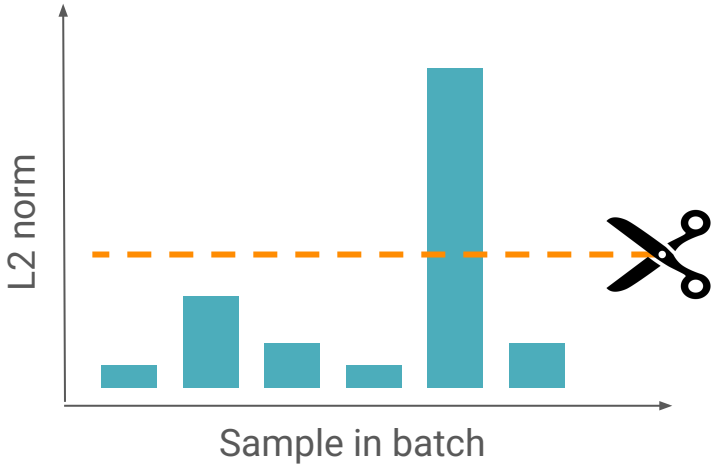


Evaluation metrics

1. Accuracy (proportion of correctly classified examples)
2. AUC (probability the model ranks positive examples higher than negatives)
3. Training time

Differentially Private SGD

Differentially private stochastic gradient descent



+ **replace BatchNorm with GroupNorm**
(uses channel stats, not batch stats)

- Enables training deep models with privacy guarantees by
- clipping gradient norms and adding Gaussian noise

Key privacy parameters

ε (epsilon) — privacy budget

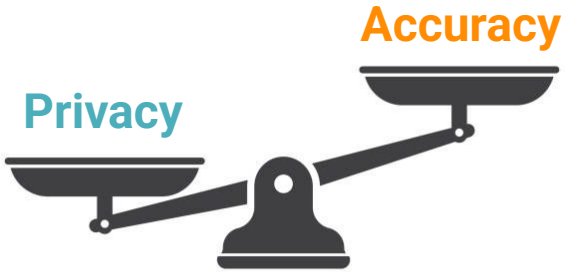
δ (delta) — failure probability

clipping threshold (C) — controls sensitivity

noise multiplier — amount of added noise

Challenges of DP-SGD in deep learning

If gradients are clipped too much and/or too much noise is added it may **significantly** degrade accuracy.



Any decision about the model based on the training dataset makes it vulnerable to privacy attacks.

➡ Privacy budget ε must cover model training + hyperparameter search

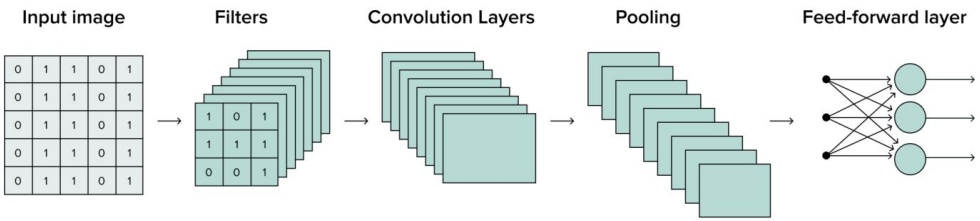
Solution Approach

Solution approach

Target privacy: $\varepsilon = 2$[1], $\delta = 1/(\text{train size})^2$

Compact CNN (~230K parameters)

GroupNorm instead of BatchNorm



+ Opacus

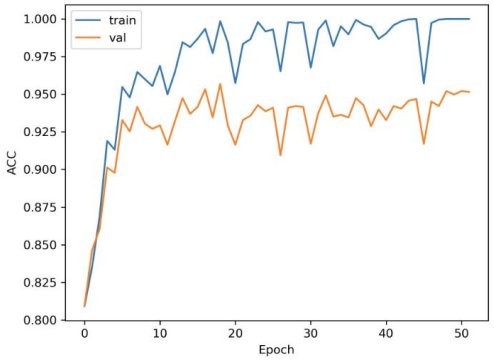
No privacy budget spent on tuning

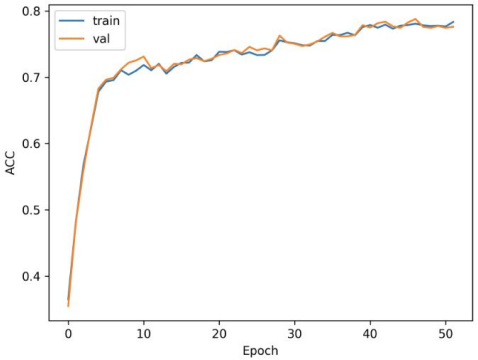
- Used **DermaMNIST** (public, 6 classes) for hyperparameter tuning
- Clipping = median gradient norm during training

[1] Rahman, Md Atiqur, et al. "Membership inference attack against differentially private deep learning model." Trans. Data Priv. 11.1 (2018): 61-79.
Image: https://serokell.io/blog/introduction-to-convolutional-neural-networks

Results

Results: non-private vs private model





ε δ

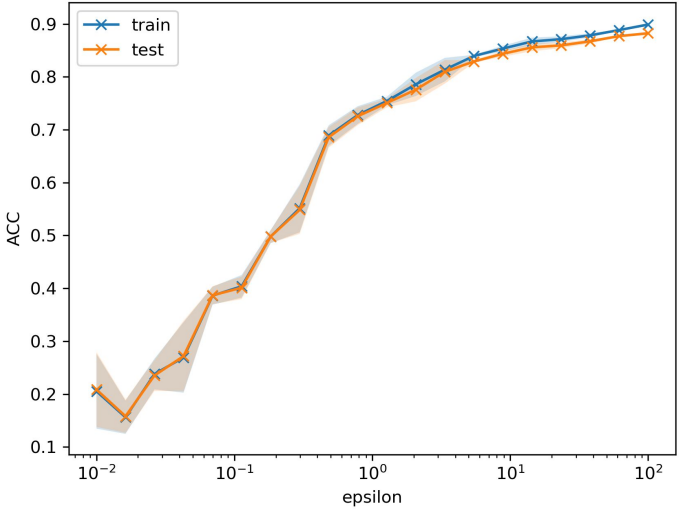
$(2\ ,\ 7 \cdot 10^{-9})$

	Non-private CNN	DP-SGD CNN
ACC	0.945 ± 0.002	0.772 ± 0.022
AUC	0.997 ± 0.000	0.952 ± 0.010
Training Time (s)	228.4 ± 15.6	360.33 ± 41.05
Number of epochs	44.3 ± 3.3	51.0 ± 7.0
Epoch time (s)	4.202 ± 0.003	5.91 ± 0.48

mean for the three weights' initializations

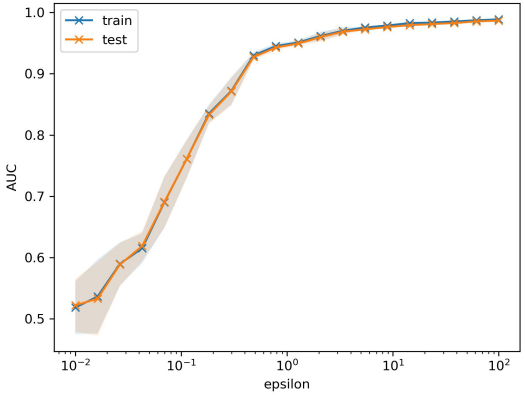
- DP-SGD training **reduced the generalization gap,** but accuracy (ACC) **decreased by 17 percentage points**
- Training time ~**1.5 times longer**
- **Higher sensitivity** to initialization (larger std deviation)

Results: effect of the privacy budget



ε < 1 → accuracy drops, variance ↑

ε > 8 → generalization gap appears
(recommended upper privacy limit for DL[2] is <10)



[2] Ponomareva, Natalia, et al. "How to dp-fy ml: A practical guide to machine learning with differential privacy." *Journal of Artificial Intelligence Research* 77 (2023)

Detour

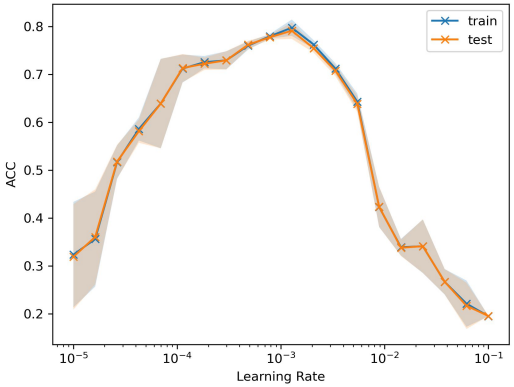
Could we have done better?

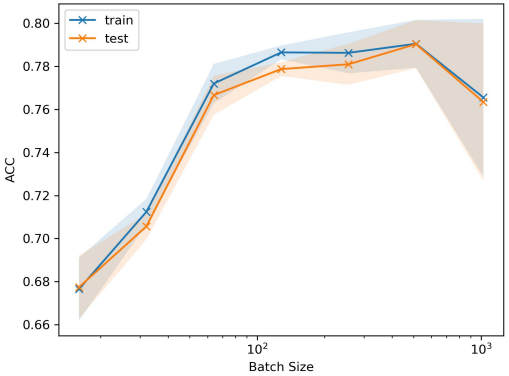
Was the accuracy limited by **poor parameter choices**?

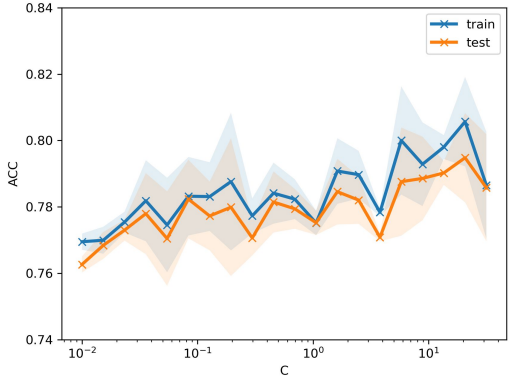
Chosen learning rate (0.001): **near optimal**

Batch size (128): **slightly below optimal**

Clipping (0.13): weak trend, high variance





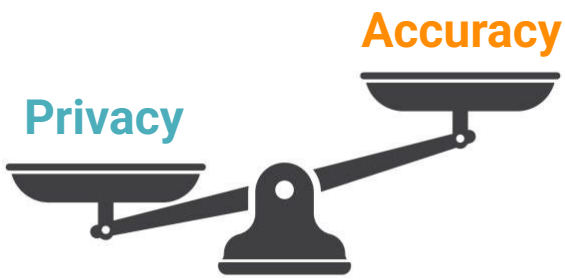


never exceeds 81%

Allocating privacy budget for tuning wouldn't help much!

Conclusions

- $\varepsilon = 2$, $\delta = 7\times10^{-9}$ can provide privacy protection of the data
- **80% accuracy** (best initialization) → not enough for decision-making
- still useful for lab support or ensembles with other models
- Training time of our private model is significantly higher **(>58%)** than its non-private version



Key takeaway

Allocating privacy budget for tuning is inefficient; using parameters tuned on a public datasets provides comparable accuracy for the compact CNN model.

Demo Time

Github repo

Thank you for your attention!